Freddy Shaikh

SENIOR SOFTWARE DEVELOPER · APPLIED AI & CLOUD INFRASTRUCTURE

Toronto, Canada

+1(437) 267-3291 | shaikhfh1@gmail.com | freddyshaikh.com | Freddy-S3 | freddy-shaikh

Summary

Senior software engineer with 7+ years of software development in one or more programming languages (Python, Go, Java, C#, TypeScript), spanning software design and architecture through testing, maintaining, and launching software products in production. Google Cloud Certified Professional Cloud Architect. Currently the designated lead for AI agent and agentic framework development at Morningstar, where I took an MCP (Model Context Protocol) server from technical design review to production release and built the developer productivity infrastructure the team now codes on by default.

Skills

AI & Agentic	AI agent frameworks (Model Context Protocol), agentic orchestration, LLM application delivery, prompt engineering, AI-assisted engineering harnesses
Languages	Python, Go, Java, C#, TypeScript, JavaScript, SQL, Bash
Infrastructure & Distributed Systems	Service decomposition, microservices, containerization (Docker, Kubernetes), event streaming (Kafka), infrastructure as code (Terraform), CI/CD, public cloud (GCP, AWS)
Practice	Software design and architecture, technical design review, build infrastructure and testing frameworks, unit/integration/API testing, Agile delivery, Tableau

Experience

Morningstar

Toronto, Canada

SENIOR FULL STACK SOFTWARE ENGINEER

January 2025 - PRESENT

- Technical SME for AI agent framework development: led the architecture of an MCP (Model Context Protocol) agent server from proposal through technical design review to production release, defining the tool interface, context scoping, and failure behaviour.
- Designated lead for AI-driven engineering acceleration. Built the team's developer productivity infrastructure - reusable skill files, scoped context and token budgets, operating-mode guardrails - and mentored senior and principal engineers onto it as their default workflow.
- Launched an AI audio product end-to-end: back-end services for LLM-driven SSML speech synthesis on managed cloud model and speech APIs (AWS Bedrock, Polly), plus the front end shipping transcript and smart-summary formats.
- Designed and deployed CI/CD pipelines with infrastructure as code, cutting deployment time for serverless and container services, and automated recurring engineering workflows in Python.

Santoku Corporation / Rimm.ai

Toronto, Canada

SOFTWARE ENGINEER

August 2019 - January 2025

- Led the decomposition of a monolithic on-premises application into independently deployable cloud services, owning the service boundaries and the design trade-offs across data consistency, latency, and failure isolation.
- Built out the resulting large-scale infrastructure on public cloud (AWS) with Docker and Kubernetes orchestration, improving uptime through independent deployability and fault isolation.
- Primary technical decision-maker for custom haptic VR training systems delivered to enterprise clients: requirements, software design and architecture, implementation, automated unit/integration/API testing, launch, and five and a half years of production maintenance.

Japan Exchange Teaching (JET) Program

Tokyo, Japan

ENGLISH LANGUAGE TEACHER

August 2018 - August 2019

- Taught English to 1,500+ students in Tokyo, working in self-taught Japanese.

Projects

Agentic Engineering Harness - open source (GitHub)

- Provider-neutral framework for autonomous coding agents across GitHub Copilot, Claude Code, and Codex: one skill format projected into each host by an idempotent installer, operating modes separating supervised from unattended posture, a recoverability-over-approval model for overnight runs, and a usage-limit-aware queue that resumes work after rate-limit resets.
- Concurrency and recovery model for parallel agents: exclusive working-tree claims with heartbeats and stale-claim takeover, git-worktree isolation, and compare-and-swap writes to shared state - enforced by a pre-commit hook that rejects an unclaimed agent write, with task state journaled before each risky step so an interrupted run resumes deterministically.

Portfolio Site & Resume Pipeline - (freddyshaikh.com)

- Single-source-of-truth build pipeline: a Python generator cascades one LaTeX source into the site, PDF, and job-board exports, eliminating manual re-sync; verified in CI with an end-to-end Playwright test suite.

Certifications & Education

2024 **Google Cloud Certified: Professional Cloud Architect**, Google Cloud Certification *Toronto, Canada*

2023 **AWS Certified Solutions Architect - Professional**, AWS Training and Certification *Toronto, Canada*

2026, 2025 **AWS Certified Machine Learning Engineer - Associate; AWS Certified AI Practitioner**, AWS Training and Certification *Toronto, Canada*

2018 **Bachelor of Honours Science**, McMaster University *Hamilton, Canada*